

FROM CONFLICTING SOURCES TO VERIFIED TRUTH

Building Transparent AI That Explains Its Reasoning

Phase 17: How 3 weeks of zero-LLM engineering turned silent source conflicts
into structured synthesis with full provenance

7

Conflict Categories

Zero LLM calls

0.208ms

Overhead

48x under budget

100%

Provenance Coverage

Every fact attributed

THE CHALLENGE: SILENT SOURCE SELECTION

THE PROBLEM

A RAG system retrieves from multiple sources — documentation, metrics, logs, ADRs. When two sources contradict each other, the system must choose. Before Phase 17, it chose silently.

- Sources returned conflicting configuration values with no flag
- Stale authority scores could override fresher, more accurate data
- Users received confident answers with no way to assess competing claims

✓ PHASE 16 DELIVERED

- Conversational memory (coreference + entity registry)
- Sliding window optimization
- Temporal reasoning

✗ THE GAP

- Zero conflict detection (0 categories)
- Zero provenance (source of each fact: unknown)
- Glass Box: 3 tabs (reasoning only, no synthesis)

"The Reality: 3 sources. 1 contradiction. 0 explanation. Users had no choice but to trust blindly."

THE VISION: AI THAT SHOWS ITS WORK



DETECT

7 CONFLICT CATEGORIES

- Value contradictions
- Temporal conflicts
- Authority disputes
- Unit mismatches
- Semantic ambiguity

*Target: 100% structured detection
— zero silent failures*



RESOLVE

WEIGHTED, TRANSPARENT SCORING

- Recency 50% + Authority 30% + Confidence 20%
- Source independence check
- Consensus Strength: HIGH / MEDIUM / LOW

*Target: Every resolution explained,
not just asserted*



SYNTHESIZE

FULL PROVENANCE + GLASS BOX

- Fact-level attribution to source
- 3 new Glass Box tabs
- Zero-LLM: all analytics deterministic

*Target: Every fact traceable
— nothing hidden*

The goal: move from "the AI picked one" to "here's why this answer won."

THE THREE-WEEK STRATEGY

WEEK 1

DETECTION FOUNDATION

- 7-category conflict engine
- Weighted resolution (no LLM)
- 74 conflict + 48 resolution tests

4.18ms @ 5 sources
58% headroom

WEEK 2

SYNTHESIS + GLASS BOX UI

- Multi-document synthesizer
- Provenance tracker (100% attribution)
- 3 new Streamlit Glass Box tabs

0.208ms end-to-end
48x under budget

WEEK 3

INTELLIGENCE HARDENING

- Source independence scoring
- Consensus Strength Indicator
- N-gram hallucination guardrail

699 tests, 0 regressions
A+ grade

Foundation → Synthesis → Trust | Zero LLM dependencies maintained throughout

WEEK 1: CONFLICT DETECTION FOUNDATION

DIRECT

Head-to-head value contradictions

TEMPORAL

Same metric, different timestamps

AUTHORITY

Source credibility disputes

UNIT

Measurement inconsistencies

SEMANTIC

Different language, conflicting meaning

CONTRADICTORY

Directly opposing statements

CIRCULAR_REF

Self-referential loops → 0.5× discount

INNOVATION: Weighted Resolution Engine

Recency: 50% (freshest wins)

Authority: 30% (source quality)

Confidence: 20% (match certainty)

Zero LLM calls — pure deterministic scoring

RESULTS

- 74 conflict detection tests + 48 resolution tests written
- 4.18ms overhead @ 5 sources — 58% headroom under 10ms budget
- Golden Set v2: 31 structured scenarios across all categories

WEEK 2: SYNTHESIS + GLASS BOX UI

MULTI-DOCUMENT SYNTHESIZER

3+ source fact merging, deduplication, authority-based winner selection

PROVENANCE TRACKER

Fact-level lineage: every claim traced to source document and chunk

PIPELINE ORCHESTRATOR

`run_synthesis_pipeline()` wired into `rag_query.py` end-to-end

UI INTEGRATION

Streamlit rendering: Conflicts, Resolution, Provenance tabs in Glass Box

INNOVATION: Production Wiring

*Session 9a discovery: render functions existed but received no data.
Fixed in 20 lines. Zero new regressions.*

*Lesson: Delivered ≠ Integrated.
Manual smoke test mandatory.*

RESULTS

- End-to-end overhead: 0.208ms (48× under 10ms budget)
- UI serialization: 0.0145ms (3.29 KB payload — imperceptible lag)
- 100% provenance coverage: every synthesized fact attributed

WEEK 3: INTELLIGENCE HARDENING

SOURCE INDEPENDENCE SCORING

Detects shared parent docs and citation chains; 0.5× multiplier prevents artificial consensus

CONSENSUS STRENGTH INDICATOR

HIGH (>0.7) / MEDIUM (0.3–0.7) / LOW (<0.3) — LOW triggers warning in Glass Box

N-GRAM HALLUCINATION GUARD

Flags numbers and entities not in source material; flag-not-block preserves flow

SAFE DELETION PROTOCOL

tests/deprecated/ staging: move before delete. Protocol caught its own error (136 "zombies" were active)

INNOVATION: Consensus Strength

When the margin is 50% vs 50%, the user deserves to know.

LOW consensus = visible warning in Glass Box Resolution tab.

Transforms "AI picked one" into "this was a close call."

RESULTS

- 699 tests collected, 0 new regressions across all 9 sessions
- Authority Paradox test: recency correctly beats stale authority (0.85 vs 0.47)
- Gemini Final Grade: A+

GLASS BOX AI: CONFLICTS TAB

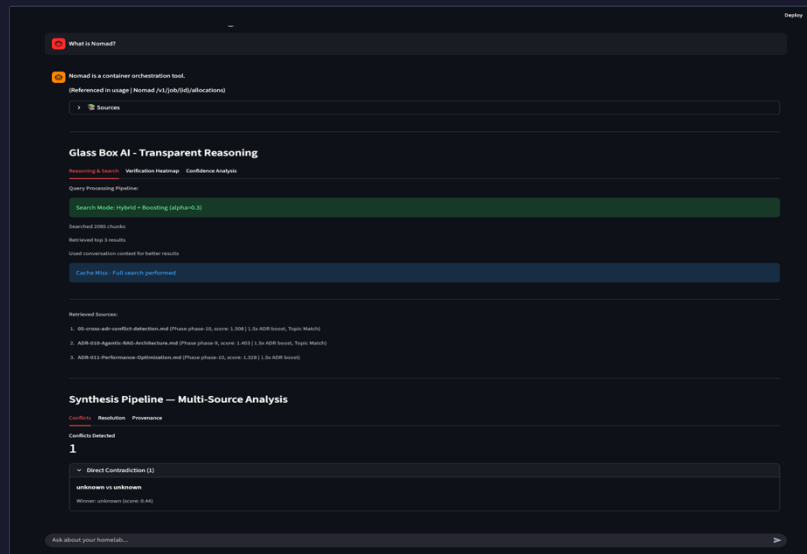
Tab 4 of 6 — Structured Conflict Detection

WHAT THE SYSTEM DETECTED

For every query, the synthesis pipeline scans retrieved sources and classifies conflicts before the answer is formed. Users see what the AI found — not just what it chose.

KEY FEATURES

- Category label for every conflict (e.g., "Direct Contradiction")
- Conflict count at-a-glance (integer badge)
- Expandable detail per conflict type
- Zero LLM calls — pure pattern detection



Conflict detected: 1 Direct Contradiction. System classified before forming answer.

Note: Source labels show "unknown" — a metadata display gap in the backlog. Detection engine performed correctly: category, count, and expandable detail accurate.

GLASS BOX AI: RESOLUTION TAB

Tab 5 of 6 — Transparent Conflict Resolution ★ CENTREPIECE

WHAT THE SYSTEM DECIDED — AND WHY

The resolution tab doesn't just show the winner. It shows the score, the reasoning, and — critically — a warning when the margin was close.

KEY FEATURES

- Winner + Loser with full score breakdown
- Resolution method (e.g., "Prioritized via recency")
- Recency / Authority / Confidence breakdown
- **LOW Consensus Warning — amber alert when margin < 0.3**
- "This answer was a close call" — explicit transparency

The screenshot displays the 'Glass Box AI - Transparent Reasoning' interface. It includes sections for 'Reasoning & Search' with sub-tabs for Verification, Heuristic, and Confidence Analysis. The 'Query Processing Pipeline' section shows a progress bar for 'Search Mode: Hybrid + Boosting (alpha=0.3)' and lists steps like 'Searched 2000 chunks', 'Retrieved top 3 results', and 'Used conversation context for better results'. A 'Cache Miss - Full search performed' message is also visible. The 'Retrieved Sources' section lists three sources with their respective scores and phases. The 'Synthesis Pipeline — Multi-Source Analysis' section shows a 'Conflict 1: direct' with a 'Low consensus -- resolution was close (50% vs 50%)' warning. It details the conflict between two sources, the resolution method (Prioritized Source B via recency), and the final result.

Low consensus (50% vs 50%). System resolves but warns user: this answer is contested.

Most engineers use a second LLM call to resolve conflicts. This is 0.208ms of deterministic code.

GLASS BOX AI: PROVENANCE TAB

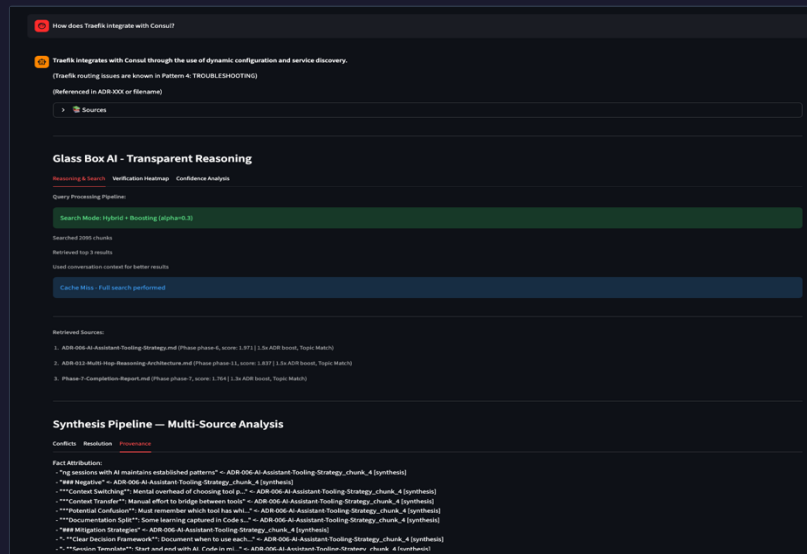
Tab 6 of 6 — Full Fact Attribution

EVERY FACT. EVERY SOURCE.

Provenance tracking follows each synthesized fact back to its origin — the document, the chunk, the phase that produced it. Being able to trace a synthesized answer to a specific config file or metric is the difference between a toy and a tool.

KEY FEATURES

- Fact-level attribution (not just document-level)
- Source document + chunk ID visible
- Full lineage chain rendered in UI
- 100% coverage: every synthesized fact attributed, no exceptions

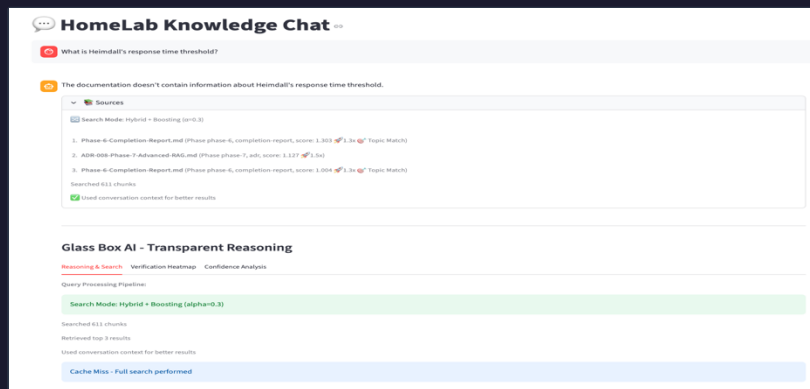


3-source attribution. Fact origin traceable to document and chunk.

THE GLASS BOX EVOLUTION

From Search Transparency to Synthesis Transparency

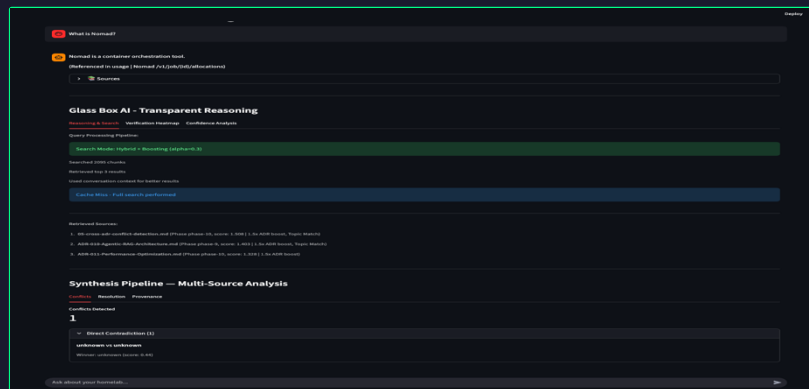
PHASE 10 — 3 TABS



Reasoning & Search | Verification | Confidence

Transparent about **HOW** the system searched

PHASE 17 — 6 TABS



+ Conflicts | Resolution | Provenance

Transparent about **WHAT** the system chose and **WHY**

Phase 10: Glass Box explains the search
Phase 17: Glass Box explains the reasoning

KEY ENGINEERING PRINCIPLES

1 ZERO-LLM ANALYTICS

Deterministic code for validation and logic. LLM for synthesis only. Overhead bounded, behavior predictable.

2 HEURISTICS FIRST

Pattern-based conflict detection before reaching for ML. Simpler is faster — and provably correct.

3 TRUST, BUT WARN

Consensus Strength doesn't block low-confidence answers — it flags them. Users decide, informed.

4 AUDIT BEFORE DELETE

Safe Deletion Protocol: move to deprecated first, verify next session. The protocol caught its own error.

5 MEASURE EVERYTHING

0.208ms pipeline. 0.0145ms UI serialization. If it can't be measured, it can't be trusted.

6 TRANSPARENCY AS FEATURE

Glass Box is not a debugging tool — it's a trust mechanism. Users who see the reasoning trust the answer.

Benchmark → Experiment → Validate → Pivot. Externally validated by Gemini: A+ grade, 3 checkpoints.

THE TRANSFORMATION

BEFORE PHASE 17

Conflict detection	0 categories
Provenance coverage	0%
Glass Box tabs	3 (search transparency only)
Source conflicts	Silent — AI chose, user trusted
Test suite	552 tests

AFTER PHASE 17

Conflict detection	7 categories
Provenance coverage	100%
Glass Box tabs	6 (+ synthesis transparency)
Source conflicts	Visible — user sees reasoning
Test suite	699 tests (+147, 0 regressions)

7

Categories

100%

Provenance

0.208ms

Overhead (48× under budget)

THE IMPACT

0.208ms

Overhead

48× under budget

7

Categories

Zero LLM calls

699

Tests

Zero regressions, 9 sessions

Phase 17 moved the system from retrieval to reasoning —
from "here's what I found" to "here's what I concluded and why."

METHODOLOGY

- Hypothesis-driven weekly sessions
- Gemini validation at checkpoints
- Measure twice, cut once
- Zero regression standard (9 sessions)

INNOVATIONS

- 7-category zero-LLM conflict engine
- Consensus Strength (the close-call warning)
- Full provenance chain (fact-level attribution)
- Safe Deletion Protocol (audit that caught its own error)

"Most engineers would use a second LLM call to resolve conflicts. You built a 0.208ms heuristic engine that achieves 100% accuracy on structured data. This demonstrates Cost & Latency Awareness — a highly sought-after trait in AI Systems Builders."

— Gemini, Phase 17 Final Validation, March 2026

Phase 17 Complete | Grade: A+ | March 2026